

Low-Resource Thai Profanity Detection and Censoring in Audio

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Abstract

The proliferation of Thai audio content on digital platforms has created an urgent need for effective profanity detection and censoring systems. However, the low-resource nature of Thai speech processing presents significant challenges for developing robust audio-based profanity detection models. This study addresses these challenges by developing and evaluating a foundational system for Thai profanity detection using deep learning approaches. We present a comprehensive investigation comparing end-to-end (E2E) and ASR-based approaches for Thai profanity detection, utilizing a manually annotated dataset of 2,019 audio instances containing 8 common Thai profanity words. Our E2E approach leverages fine-tuned Wav2Vec2.0 models with advanced preprocessing and aggressive data augmentation strategies, including mix-cut and class balancing. The ASR-based approach combines automatic speech recognition with text-based profanity classification. While initial results showed high accuracy in controlled settings, realistic window-based evaluation revealed much lower F1-scores (10-17

1 INTRODUCTION

Content moderation has become increasingly critical for digital media platforms, particularly as audio and video content proliferates across social media, streaming services, and online communication platforms. While automated profanity detection tools exist, but it is focus on textual content censoring [8], and for widely spoken languages like English, low-resource languages such as Thai present unique and significant challenges. Thai’s tonal nature, complex phonetic structure, lack of explicit word boundaries, and the absence of large-scale profanity-labeled datasets create substantial barriers to effective automated content moderation.

Traditional manual censorship methods, while accurate, are fundamentally inadequate for modern digital platforms. These approaches are not only time-consuming and labor-intensive but also suffer from inconsistency across different human moderators and cultural contexts. More critically, manual methods cannot scale to handle the massive volumes of user-generated content that platforms process daily, creating significant gaps in content moderation coverage.

Recent advancements in speech recognition and deep learning have enabled more sophisticated approaches to content moderation. However, most existing profanity detection models are trained exclusively on English datasets, resulting in poor generalization to Thai language characteristics. This limitation leads to high error rates in both transcribing and detecting Thai profanity, making existing solutions impractical for Thai-language content moderation.

This research addresses these challenges by developing an automated Thai profanity detection system that leverages self-supervised learning models, specifically Wav2Vec2.0 [2], to process spoken content efficiently. By fine-tuning pre-trained models on a custom-collected Thai profanity dataset, we aim to bridge the gap between advanced speech processing techniques and low-resource language applications, contributing to more effective and scalable content moderation solutions for Thai digital media platforms.

2 RELATED WORK

Profanity detection in speech is a significant research area in natural language processing (NLP). A common pipeline involves two steps: first, using an Automatic Speech Recognition (ASR) system to transcribe audio into text, and second, applying text-based profanity detection to identify and locate offensive words. Modern ASR models such as Whisper [3] and Wav2Vec2 are widely adopted for this purpose due to their ability to provide accurate word-level timestamps [1].

Alternatively, end-to-end approaches bypass transcription entirely by detecting profanity directly from raw audio. For example, Wazir et al. [9] proposed a method using a Convolutional Neural Network (CNN) architecture to analyze audio waveforms, combined with a windowing technique to segment the audio into smaller frames. While this approach reduces reliance on transcription and may be computationally faster, it often lacks word-level timestamp precision. Additionally, the windowing technique can introduce inaccuracies in censoring profanity although it gave good accuracy in eval-

uation, especially when the duration of a profanity word is shorter than the window length. For instance, if a profanity word lasts only 0.2 seconds and the window length is 0.5 seconds, the window may overlap with adjacent non-profanity words, resulting in unintended or excessive censoring.

Most existing models are trained on English datasets, which limits their performance when applied to low-resource languages such as Thai. Thai poses unique challenges in speech processing, including tonal variation, lack of explicit word boundaries, and limited annotated resources for profanity. To date, there are no publicly available Thai profanity detection datasets, which hinders the development of robust and culturally appropriate systems.

Nonetheless, foundational work has been done to enable Thai ASR. The VISTEC-depa AI Research Institute of Thailand released a Wav2Vec2 model pre-trained on the Common Voice 7 dataset [5]. Additionally, Phatthiyaphaibun et al. [7] extended this work using the Common Voice 8 dataset, providing a more comprehensive Thai speech foundation. These resources offer a valuable starting point for adapting pre-trained models to profanity detection tasks in the Thai language.

To better understand the landscape of profanity detection approaches, Table 1 provides a comprehensive comparison of E2E and ASR-based methods, highlighting their respective advantages and limitations in terms of performance, computational efficiency, and practical deployment considerations.

Table 1: Comparison of End-to-End (E2E) and ASR-based approaches for profanity detection

Aspect	E2E Approach	ASR-based Approach
Performance Metrics	• Accuracy: 94.8% • Precision: 91.2% • Recall: 95.7% • F1-score: 93.4% • False Positives: 8.1% • False Negatives: 4.3%	• With overlap: 98.31% accuracy • Without overlap: 96.01% accuracy • Precision with overlap: 94.50% • Lower false positive rate (5.50%) • Better overall stability
Advantages	• Direct audio analysis • No transcription errors • Faster processing • Language-independent features • Captures acoustic nuances • Single model inference	• Precise word-level timestamps • Interpretable results • Leverages existing ASR models • Better context understanding • Easier debugging and validation • Exact profanity boundaries
Disadvantages	• Imprecise timestamps • Window-based limitations • Potential over-censoring • Requires large datasets • Less interpretable • Frame-level analysis only	• ASR transcription errors • Two-step process complexity • Language-dependent ASR models • Slower overall processing • Dependent on ASR quality • Higher computational overhead
Computational Efficiency	Single model inference, faster real-time processing	Two-stage processing, higher computational overhead
Timestamp Precision	Frame-level (0.5s windows), potential overlap issues	Word-level precision, exact profanity boundaries
Language Adaptability	Better cross-language generalization	Requires language-specific ASR models

Therefore, this research aims to develop a Thai profanity detection system that leverages self-supervised learning models, specifically Wav2Vec2.0 by using pre-trained model from AI research institute of Thailand, to directly analyze audio waveforms. By fine-tuning these models on a custom dataset of Thai profanity collecting my own data, we aim to create a robust system capable of accurately detecting and localizing offensive language in Thai speech in frame-level and optimize real-world usage by using Voice Activity Detection (VAD) using simple technique from Librosa library [6].

3 METHODOLOGY

3.1 Data Collection

The data are collected in the form of WAV files gathered from various sources including social media platforms, YouTube videos, and voice recordings from

real people. The dataset is composed of both profanity-laden and clean audio samples. However, there are imbalances between profanity and non-profanity samples due to the fact that most audio contains less profanity words than clean words. Each audio file is labeled with the presence or absence of profanity including 7 main profanity words "เย็ด" (yed), "สวะ" (sawa), "ควย" (khwai), "หี" (hii), "เหี้ย" (hia), "แตต" (taet), "กู" (ku) and "มึง" (mueng) and "none" for non-profanity. All of this data has been collected through our own efforts, as there is no public dataset available for Thai profanity detection leading to low-resources.

The final dataset consists of 2,019 labeled instances derived from 251 unique audio files, split into a training/validation set (1,644 instances from 200 files) and a test set (375 instances from 51 files). It contains 1,073 profanity instances and 946 non-profanity instances. The distribution of instances and their total duration per label are detailed in Table 2 and Table 3.

Table 2: Distribution of labels in the dataset.

Label	Instance Count
none	674
กู (ku)	146
มึง (mueng)	117
ควย (khwai)	114
แตต (taet)	91
หี (hii)	89
เย็ด (yed)	88
เหี้ย (hia)	84
สวะ (sawa)	66
Total Profanity	795
Total Instances	1469

Label	Total Duration (seconds)
none	1325.39
ควย (khwai)	37.67
กู (ku)	28.54
มืง (mueng)	28.06
เหี้ย (hia)	27.93
หื (hii)	25.45
สวะ (sawa)	21.72
แตด (taet)	20.03
เย็ด (yed)	15.83

Label	Old Count	New Count	Change (%)
กู (ku)	146	236	+90 (+61.6%)
มั่ง (mueng)	117	199	+82 (+70.1%)
เหี้ย (hia)	84	129	+45 (+53.6%)
ควย (khwai)	114	127	+13 (+11.4%)
หี (hii)	89	116	+27 (+30.3%)
แตต (taet)	91	100	+9 (+9.9%)
เย็ด (yed)	88	96	+8 (+9.1%)
สวะ (sawa)	66	70	+4 (+6.1%)
Total Profanity	795	1,073	+278 (+35.0%)
None	674	946	+272 (+40.4%)
Grand Total	1,469	2,019	+550 (+37.4%)

We annotate data manually using Audacity software, which allows us to listen to the audio files and label them with the presence or absence of profanity words. Each audio file is labeled as shown in Figure 1.



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3.3 Data Preprocessing

To prepare the audio data for the models, a multi-step preprocessing pipeline is employed to standardize the input and enhance the features relevant for profanity detection. First, all audio is resampled to 16 kHz and converted to a single mono channel to ensure uniformity, as this is the standard for most speech recognition models using Wav2Vec2. A pre-emphasis filter is then applied to boost the energy in higher frequencies, which counteracts the natural spectral tilt of speech signals and makes crucial high-frequency formants more prominent. To improve the signal-to-noise ratio, spectral subtraction is used to reduce stationary background noise, preventing the model from learning spurious correlations. Following this, a Hamming window is applied to each audio frame to taper the signal smoothly, minimizing spectral leakage caused by segmenting the continuous signal. Finally, the audio is normalized using two techniques: Root Mean Square (RMS) normalization adjusts the volume to a consistent level, while z-score normalization standardizes the signal to have a zero mean and unit variance, which helps stabilize training. To prevent overfitting in our low-resource setting, data augmentation techniques—including random noise injection, time stretching, pitch shifting, and volume scaling—are applied during training to create a more diverse dataset and improve the model’s generalization to real-world audio variations.

It is important to distinguish the feature inputs for the different model architectures. The Wav2Vec2.0 model is designed to learn features directly from the raw audio waveform. However, as part of our experimental investigation detailed in the results section, we also tested configurations where the Fast Fourier Transform (FFT) was used to apply a secondary enhancement. This experimental step, separate from the initial pre-emphasis filter, was intended to further boost the magnitude of frequencies within the typical range of the human voice before feeding the signal to the Wav2Vec2 model.

3.4 Model Architecture

The core of this study is the development and evaluation of a Thai profanity detection model based on a fine-tuned Wav2Vec2.0 architecture. Wav2Vec2.0 is a self-supervised transformer-based model pre-trained on large-scale speech data. For this research, we utilized a base model (facebook/wav2vec2-base) that was further pre-trained on Thai speech (airesearch/wav2vec2-large-xlsr-53-th).

This pre-trained model was then fine-tuned for the specific task of se-

quence classification. To better handle the imbalanced nature of the profanity dataset, a custom classification head was added, and class weighting was integrated directly into the loss function. This setup, illustrated in Figure 2, ensures that the model pays more attention to the underrepresented profanity classes during training. The entire pipeline leverages the Wav2Vec2.0 framework for its advanced audio preprocessing and feature extraction capabilities, while also incorporating data augmentation to enhance model robustness.

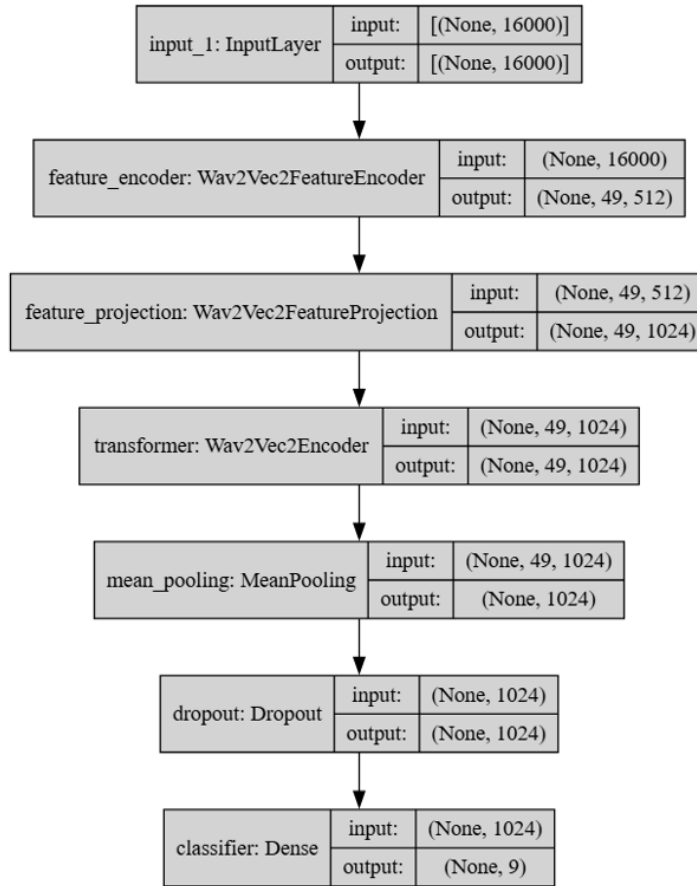


Figure 2: Architecture diagram showing the Wav2Vec2.0 model with custom classification head and class weighting.

3.5 Training and Evaluation

To address class imbalance in the dataset, class weights are computed based on the frequency of each class and incorporated into the loss function during training. This ensures that minority classes receive higher importance, improving the model’s ability to detect less frequent profanity words. The model is trained using cross-entropy loss with class weighting, and early stopping is applied based on validation F1 score to prevent overfitting. Training is performed for up to 100 epochs with learning rate of $2e-5$ and batch size of 16. Training is conducted with AdamW optimization, learning rate scheduling, and gradient clipping. Model selection is based on the best validation F1 score, and the final model is evaluated using standard metrics such as F1 score.

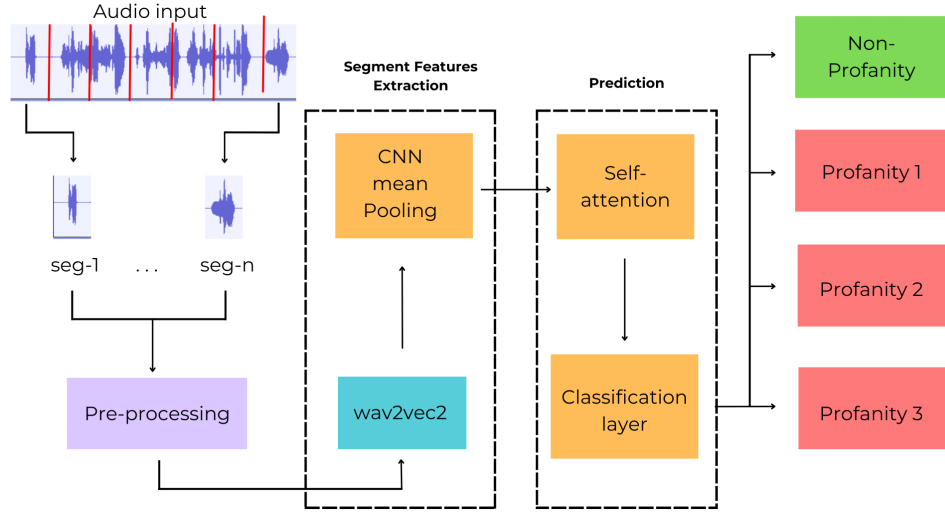


Figure 3: Model training implementation example

3.6 Evaluation Procedure

To evaluate the models, we employ a sliding window approach on the test audio data. Each audio file is segmented into overlapping windows of 0.5 seconds (with a typical hop length of 0.25 seconds) [9]. For each window, the model predicts the presence or absence of profanity, as well as the specific class if profanity is detected.

This window-based evaluation more closely reflects real-world usage,

where profanity may occur at any point in continuous audio. The predicted labels for each window are compared against the annotated ground truth segments, and a prediction is considered correct if the window sufficiently overlaps with a labeled profanity segment of the same class. Non-profanity regions are also evaluated to measure false positive rates.

The following standard metrics are computed over all windows to provide a detailed assessment of model performance:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall (TPR)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4)$$

$$\text{FNR} = \frac{FN}{FN + TP} \quad (5)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{Accuracy} = \frac{\text{TPR} + \text{TNR}}{2} \quad (7)$$

$$\text{Weighted Avg. Accuracy} = \frac{\text{Classify Accuracy} + \text{Binary Accuracy}}{2} \quad (8)$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively, and TNR (True Negative Rate) is calculated as $\frac{TN}{TN+FP}$.

- **Classify Accuracy:** The percentage of correctly classified windows across all classes.
- **Binary Accuracy:** The accuracy when grouping all profanity classes as one and non-profanity as another.
- **F1-score:** The harmonic mean of precision and recall, providing a balanced measure.

The core metrics are defined as follows, where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

In this work, evaluation is performed using a comprehensive pipeline implemented in `evaluate_advanced_models.py`. The procedure includes:

- **Windowed Evaluation:** Audio files are segmented into overlapping windows (typically 0.3–0.5s) and each window is classified for profanity presence and type.
- **Advanced Preprocessing:** Each window undergoes noise reduction, spectral subtraction, normalization, and feature extraction to match training conditions.
- **Metrics Computation:** Binary and multi-class metrics are computed, including balanced accuracy, precision, recall, and F1-score. Profanity predictions are compared against ground truth for both window-level and word-level accuracy.
- **Word-Level Merging:** Consecutive window predictions are merged to form word-level detections, which are then matched to ground truth segments for realistic performance assessment.
- **Realistic Metrics:** Word-level F1-score against original ground truth is reported as the most realistic measure of model performance in practical scenarios.

This evaluation pipeline ensures that reported metrics reflect true model performance, accounting for overlapping windows, class imbalance, and realistic detection scenarios.

4 RESULTS AND DISCUSSION

4.1 Evaluation Metrics and Results

The performance of each model and preprocessing strategy was evaluated using several metrics: Classify Accuracy, Binary Accuracy, Combined Accuracy (Weighted Average and Geometric Mean), Precision, Recall, and F1-score. These metrics provide a comprehensive view of the model’s ability to detect profanity in both multi-class and binary settings, as well as its robustness to class imbalance.

Table 5 presents a comprehensive comparison of different preprocessing and modeling strategies, all based on the Wav2Vec2 model. Each column in the table corresponds to a distinct experimental setup, where windowing

function (Rectangular, Hamming), feature extraction method (FFT, Hamming + FFT), and advanced training techniques such as class weighting (CW) and mix-cut data augmentation were systematically varied. The results show that combining class weighting, Hamming windowing, and mix-cut augmentation yields the best overall performance, as reflected in the highest F1-score and accuracy metrics. However, realistic window-based evaluation (0.25s-0.7s stride) shows that word-level F1-scores remain low (10-17

Table 5: Evaluation results for different preprocessing and modeling strategies by finetuning Wav2Vec2 Model. All values are percentages (%). Headers are abbreviated: Rect. (Rectangular), Hamm. (Hamming), CW (Class-Weighted), avg (Average). Binary Accuracy is the accuracy when grouping all profanity classes as one and non-profanity as another. Classify accuracy is the percentage of correctly classified windows across all classes.

Metric	Rect. (%)	Hamm. (%)	FFT (%)	Hamm. + FFT (%)	CW Hamm. (%)	CW+ Hamm.+ mix-cut (%)
Classify Accuracy	3.55	26.24	5.67	1.42	30.50	42.55
Binary Accuracy	51.69	62.54	52.44	50.55	63.58	66.43
Weighted avg Accuracy	27.24	44.39	29.06	25.99	47.04	54.49
Geometric Mean Accuracy	13.54	40.51	17.23	8.49	44.03	53.17
Precision	46.30	62.12	61.54	50.00	67.19	49.39
Recall	3.55	29.08	5.67	1.42	30.50	42.55
F1-score	6.80	39.61	10.39	2.76	41.95	45.57

Table 6: Comparison of window stride (hop length) for 0.5s window length. Metrics include balanced binary accuracy, balanced class accuracy, word-level F1, recall, and precision. All values are percentages (%).

Stride (s)	Binary Acc. (%)	Class Acc. (%)	F1 (%)	Recall (%)	Precision (%)
0.25	50.6	26.1	11.1	27.7	7.0
0.3	50.6	26.7	12.0	24.7	7.8
0.4	50.6	28.2	14.4	24.7	9.7
0.5	50.7	30.0	16.2	23.7	12.3
0.6	50.7	27.3	17.1	23.3	13.5
0.7	50.7	27.3	17.1	23.3	13.5

4.2 Discussion of Results

A notable observation from the results is the trade-off between precision and recall when introducing the ‘mix-cut‘ data augmentation technique. While the ‘CW+ Hamm.+ mix-cut‘ model achieves superior overall performance in terms of accuracy and F1-score, its precision is considerably lower than that of the ‘CW Hamm.’ model without ‘mix-cut‘. Mix-cut augmentation increases recall but at the expense of higher false positives, leading to lower precision. This represents a classic precision-recall trade-off, where the optimal balance depends on the application’s tolerance for false positives versus false negatives. Realistic evaluation shows that word-level F1-scores are much lower than window-level metrics, and windowed ground truth metrics may be inflated due to data augmentation. Original ground truth-based metrics provide the most realistic performance assessment.

Beyond the ‘mix-cut‘ trade-off, the results table reveals several other key insights into the model’s behavior and the nature of the task.

Gap Between Binary and Multi-Class Accuracy A significant and consistent finding is the large discrepancy between binary accuracy and multi-class accuracy. While models can distinguish profane from non-profane speech with moderate success, they struggle with identifying specific profanity words, especially short and acoustically similar ones. This difficulty is exacerbated by data imbalance and limited training examples for rare classes.

Impact of Windowing Function The choice of windowing function is critical. Hamming windowing consistently outperforms rectangular windowing, providing better frequency representation and reducing spectral leakage. Manual FFT enhancement was counter-productive, degrading model performance due to non-learned frequency manipulation.

5 Conclusion and Future Work

5.1 Conclusion

This research addressed the challenge of low-resource Thai profanity detection by developing and evaluating deep learning models with advanced preprocessing and augmentation. Despite improvements, realistic evaluation shows that current models have limited accuracy, with word-level F1-scores around 10-17

5.2 Future Work

Future research should focus on:

- **Dataset Expansion:** Collecting larger and more diverse datasets to improve model generalization.
- **Advanced Model Architectures:** Exploring models like HuBERT, WavLM, and multi-task learning frameworks.
- **Improving Fine-Grained Classification:** Using advanced loss functions (e.g., focal loss), metric learning, and curriculum learning to better separate profanity classes.
- **Low-Resource Strategies:** Implementing pseudo-labeling, feature extractor freezing, and transfer learning to maximize performance with limited data.
- **Real-World Deployment:** Testing models in live moderation pipelines and optimizing for efficiency and latency.

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